Section 5.1 - Polynomials

Interval Notation: A set of real numbers that makes up a portion of the number line can be represented by an **interval**.

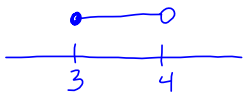
- A **closed interval** $[a,b] = a \le x \le b$, is the set of all numbers between a and b, including both endpoints.

$[3,4]$

- An **open interval**, $(a,b) = a < x < b$, is the set of all numbers between a and b, NOT including either endpoint.

$(3,4)$

- If an interval includes one endpoint, but not the other, the interval is called **half-open** or **half-closed**: $(a,b]$ or $[a,b)$



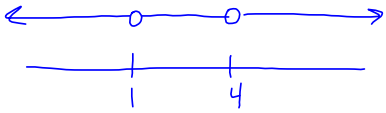
$[3,4)$

Example 1: Represent the following using interval notation.

a) $0 \le x < 9$

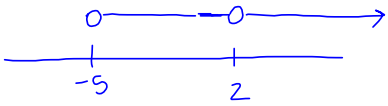
$\boxed{[0,9)}$

b) All real numbers, x, except $x = 1$ and $x = 4$



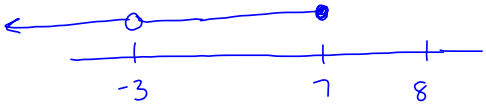
$\boxed{(-\infty,1) \cup (1,4) \cup (4,\infty)}$

c) $x > -5$ and $x \ne 2$



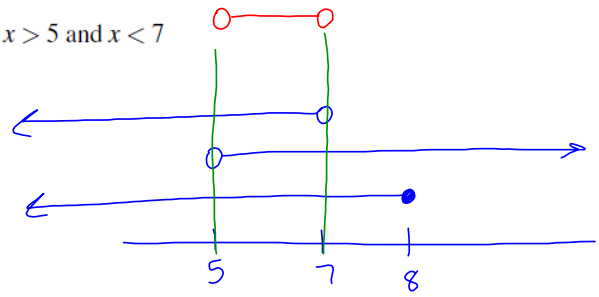
$\boxed{(-5,2) \cup (2,\infty)}$

d) $x \le 7$ and $x \ne -3, 8$



$\boxed{(-\infty,-3) \cup (-3,7]}$

e) $x \le 8$ and $x > 5$ and $x < 7$



$\boxed{(5,7)}$

Example 2: Sketch a graph for each of the following equations:

a) $2x+y=8$

b) $x^2+y^2=1$

c) $y=x^2+1$

Which of the above are functions?

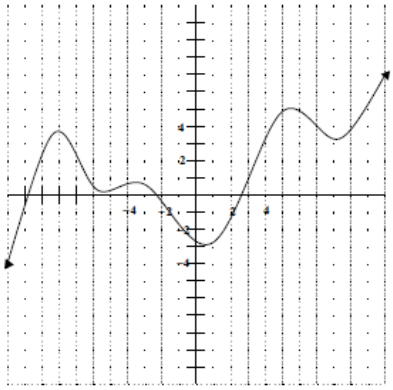
Definition: A **function** is a rule that assigns to each element x in the domain exactly _____ element in the range. The _____ is the set of all values of the independent variable (typically x) that produce real values for the dependent variable (typically y). The _____ is the set of all possible y values as x varies throughout the domain. If the correspondence is a function we use the notation $f(x)$ read "f of x". Ordered pairs of the form (x,y) can be written as $(x,f(x))$.

Example 3: Find the domain and range (using interval notation) of each function in Example 2.

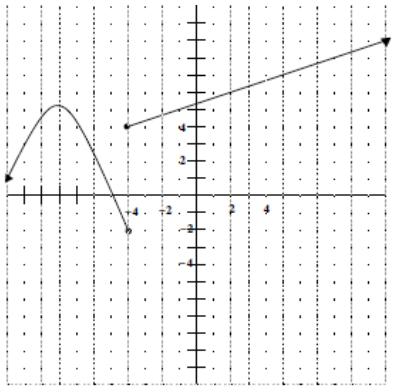
Vertical Line Test: An equation defines a _____ if each vertical line in the coordinate system passes through at most one point on the graph of the equation.

Example 4: Determine whether each is a function of x.

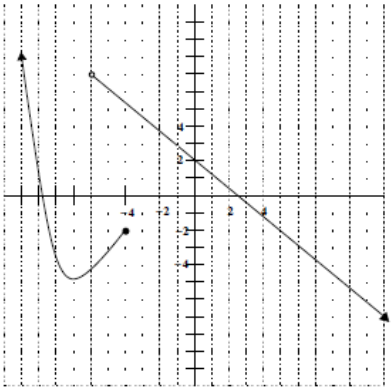
a)



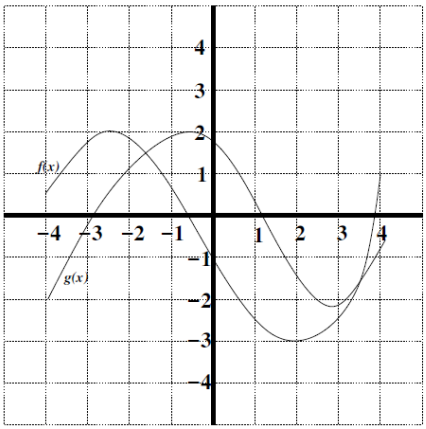
b)



c)



Example 5: Given the graph of $f(x)$ and $g(x)$ below, determine the following:



a) $f(0)$

b) $g(2)$

c) The value(s) of x for which $f(x) = 2$.

d) The approximate value(s) of x for which $f(x) = g(x)$.

Example 6: Let $f(x) = 2x^2 + x - 1$ and $g(x) = -3x^2 - x + 4$. Find (and simplify) the following:

a) $f(-1)$

b) $g(-8) - f(1)$

c) $f(a+h) - f(a)$

Example 7: Let $f(x) = 3x^2 + x$ and $g(x) = -2x^2 - 5x - 6$. Find (and simplify) the following:

a) $g(x+h) - g(x)$

$-2(x+h)^2 - 5(x+h) - 6 - (-2x^2 - 5x - 6)$

$= -2(x+h)(x+h) - \cancel{5x} - 5h - \cancel{6} + 2x^2 + \cancel{5x} + \cancel{6}$

$= -2(x^2 + 2xh + h^2) - 5h + 2x^2$

$= \cancel{-2x^2} - 4xh - 2h^2 - 5h + \cancel{2x^2} = \boxed{-4xh - 2h^2 - 5h}$

$g(2) = -2(2)^2 - 5(2) - 6$

$(x+h)(x+h)$

$= x^2 + xh + xh + h^2$

$= x^2 + 2xh + h^2$

First
Outter
Inner
Last

b) $g(a-2) \cdot f(3)$

$g(a-2) = -2(a-2)^2 - 5(a-2) - 6$

$= -2(a^2 - 4a + 4) - 5a + 10 - 6$

$= -2a^2 + 8a - 8 - 5a + 4$

$= -2a^2 + 3a - 4$

$(a-2)^2 = (a-2)(a-2)$

$= a^2 - 2a - 2a + 4$

$= a^2 - 4a + 4$

$f(3) = 3 \cdot 3^2 + 3$

$= 30$

$g(a-2) \cdot f(3) = (-2a^2 + 3a - 4)(30)$

$= \boxed{-60a^2 + 90a - 120}$

Definition: A **polynomial function** is a function that can be written in the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

for n a nonnegative integer, called the **degree** of the polynomial. The coefficients, $a_0, a_1, \ldots, a_n$ are real numbers with $a_n \neq 0$. The coefficient a_n is known as the **leading coefficient** of the polynomial.

Example 8: Which of the following are polynomials? For each polynomial, give its degree and circle the leading coefficient.

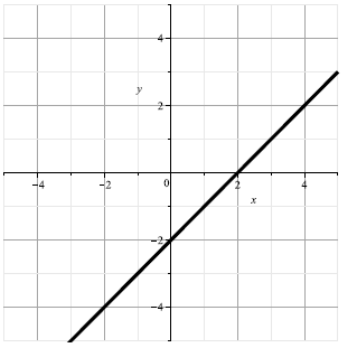
a) $f(x) = 4x^2 + 9x^7 - 5$

b) $g(x) = 20 - 3x^{1/5} - 7x^2$

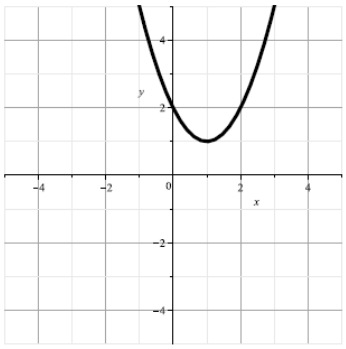
c) $h(x) = 3x^3 - 9x^{10} + 4x^7 - \pi$

d) $j(x) = 5x - \sqrt{x} + x^{-1}$

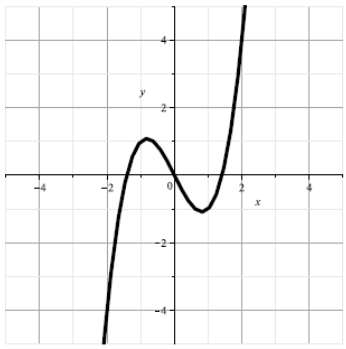
Some graphs of polynomials:



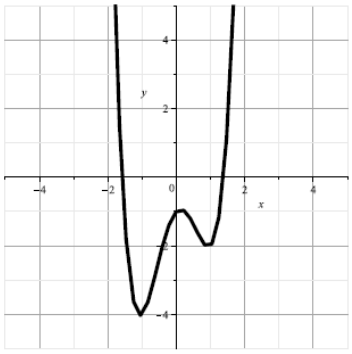
$f(x) = x - 2$



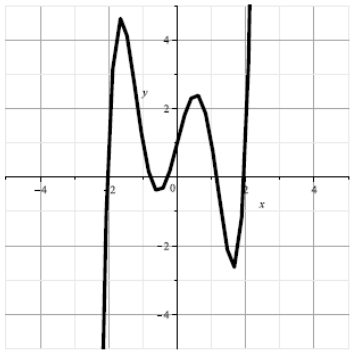
$g(x) = x^2 - 2x + 2$



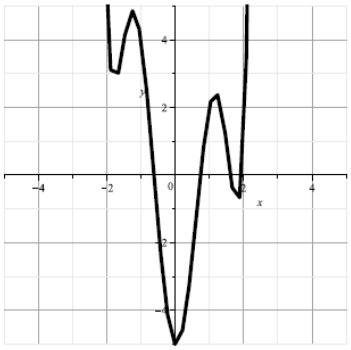
$h(x) = x^3 - 2x$



$F(x) = 2x^4 - 4x^2 + x - 1$



$G(x) = x^5 - 5x^3 + 4x + 1$



$H(x) = x^6 - 7x^4 + 14x^2 - x - 5$

Summary of Properties of Polynomials

Definition: A function of the form

is called a **quadratic function**, where a, b, and c are real numbers and $a \neq 0$. This is the **standard form**. The graph of a quadratic function is ca lled a _____.

Definition: If a, h, and k are real numbers with $a \neq 0$, then the **vertex form** of a quadratic function is

where the **vertex** (lowest or highest point on the parabola) is the point _____ where _____.

Example 9: Given $g(x) = -4x^2 + 16x - 8$,

a) Find the vertex form.

b) Find the maximum value.

c) Find the minimum value.

d) What is the domain?

e) What is the range?

Definition: The real **zeros (or roots)** of a function are its x-intercepts. To find the EXACT values of a quadratic function:

1. Factor the quadratic into two linear terms and set each equal to zero.
2. Use the quadratic formula:

Example 10: Find the EXACT location of all zeros (x-intercepts) of the following quadratic functions:

a) $f(x) = x^2 - 4$

b) $g(x) = 6x^2 - 7x - 3$

c) $h(x) = -2x^2 + 4x + 3$

d) $j(x) = x^2 + 1$

Example 11: Solve the following equations for x:

a) $x^2 + 3x = 40$

$-40 \quad -40$

$x^2 + 3x - 40 = 0$

$(x+8)(x-5) = 0$

$x+8=0 \qquad x-5=0$

$\boxed{x=-8} \qquad \boxed{x=5}$

1, 40
2, 20
4, 10
5, 8

Check:
$(x+8)(x-5)$
$= x^2 - 5x + 8x - 40$
$= x^2 + 3x - 40$

This idea only works if RHS is zero!

$\cancel{x^2 + 3x = 40}$

$\cancel{x(x+3) = 40}$

$\cancel{x = 40 \qquad x+3 = 40}$

$\cancel{x = 37}$

b) $x^3 + 2x^2 - 4x = 0$

$x(x^2 + 2x - 4) = 0$

$\boxed{x=0}$

$x^2 + 2x - 4 = 0$

$\cancel{1, 4 \leftarrow 3}$

$\cancel{2, 2 \leftarrow 0}$

Recall $ax^2 + bx + c = 0$

when $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$a = 1$
$b = 2$
$c = -4$

$x = \dfrac{-2 \pm \sqrt{2^2 - 4(1)(-4)}}{2(1)}$

$= \dfrac{-2 \pm \sqrt{20}}{2}$

$= \dfrac{-2 \pm \sqrt{4 \cdot 5}}{2} = \dfrac{-2 \pm \sqrt{4} \cdot \sqrt{5}}{2} = \dfrac{-2 \pm 2\sqrt{5}}{2} = \dfrac{\cancel{2}(-1 \pm \sqrt{5})}{\cancel{2}}$

$= -1 \pm \sqrt{5}$

$\boxed{x = 0,\ x = -1 - \sqrt{5},\ x = -1 + \sqrt{5}}$

Let's review and expand upon what we learned in Chapter 1:

Cost Function (cost of manufacturing x units of a product)

where c is the cost per unit and F is the fixed cost.

Revenue Function (revenue realized from selling x units of the product)

Profit Function (profit realized from making and selling x units of the product)

Example 12: It is found that the consumers of a particular toaster will demand 64 toaster ovens when the unit price is \$35 whereas they will demand 448 toaster ovens when the unit price is \$5. Assuming that the demand function is linear and the selling price is determined by the demand function,

a) Find the demand equation.

b) Find the revenue function.

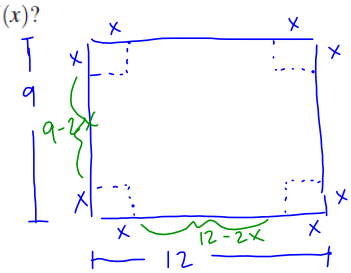
c) Find the number of items sold that will give the maximum revenue. What is the maximum revenue?

d) If the company has a fixed cost of \$1,000 and a variable cost of \$15 per toaster, find the company's linear cost function.

e) What is the company's maximum profit?

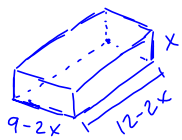
f) How many toasters should be sold for the company to break even?

Example 13: From all four courners of a 9 inch by 12 inch piece of poster board, squares are cut with dimensions x by x. The sides of the remaining pieces of poster board are turned up to form an open box. Find the volume, V, of the box as a function of x. What is the domain of $V(x)$?



volume = length · width · height

$$\boxed{V(x) = (12-2x)(9-2x)\,x}$$

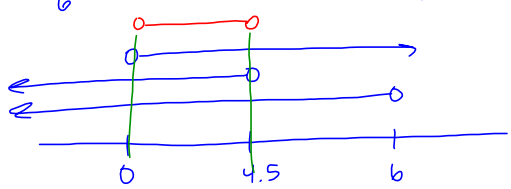


$12-2x>0$ $\quad$ $9-2x>0$ $\quad$ $x>0$

$\frac{12}{2}>\frac{2x}{2}$ $\quad$ $\frac{9}{2}>\frac{2x}{2}$

$6>x$ $\quad$ $4.5>x$

$$\boxed{(0, 4.5)}$$



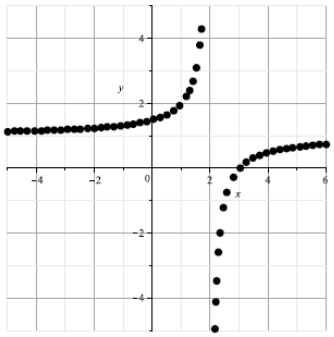
Section 5.1 Highly Suggested Homework Problems: 3, 5, 7, 9, 11, 17, 19, 23, 31, 39, 43, 47, 49, 53, 57, 67

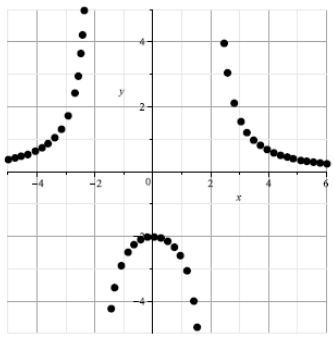
Section 5.2 - Power, Rational, and Piecewise-Defined Functions

Definition: A _____ is any function that can be written in the form

$$f(x) = \frac{n(x)}{d(x)}$$

where $n(x)$ and $d(x)$ are polynomials. The **domain** is the set of all real numbers such that $d(x) \neq 0$.





Example 1: Find the domain of each of the following functions.

a) $f(x) = \dfrac{2}{3x-5}$

b) $g(x) = \dfrac{x-7}{x^2-5x-36}$

Example 2: Find the domain of $f(x) = \dfrac{x-2}{-2x^3-4x^2+16x} = \dfrac{\cancel{(x-2)}}{-2x(x+4)\cancel{(x-2)}} = \dfrac{1}{-2x(x+4)}$, $x \neq 2$

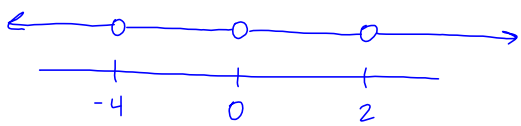
$-2x^3-4x^2+16x \neq 0$

$-2x(x^2+2x-8) \neq 0$

$\underline{-2x(x+4)(x-2)} \neq 0$

$\cancel{1,8}$

$2,4$



$-2x \neq 0$ $\quad x+4 \neq 0$ $\quad x-2 \neq 0$

$x \neq 0$ $\quad x \neq -4$ $\quad x \neq 2$

$$\boxed{(-\infty,-4) \cup (-4,0) \cup (0,2) \cup (2,\infty)}$$

Example 3: Simplify the following functions:

a) $f(x) = \dfrac{x-2}{x+4} \cdot \dfrac{x}{x+1}$

b) $g(x) = \dfrac{\dfrac{2x-1}{x+2}}{\dfrac{x-1}{3x+2}}$

c) $h(x) = \dfrac{x+7}{x-8} - \dfrac{x-2}{x+4}$

Example 4: Simplify $k(x) = \dfrac{2x-1}{x^2-2x-8} - \dfrac{x+6}{x^2-x-12}$

$$= \frac{2x-1}{(x+2)(x-4)} - \frac{x+6}{(x+3)(x-4)}$$

$$= \frac{(x+3)}{(x+3)} \cdot \frac{2x-1}{(x+2)(x-4)} - \frac{x+6}{(x+3)(x-4)} \cdot \frac{(x+2)}{(x+2)}$$

$$= \frac{(x+3)(2x-1) - (x+6)(x+2)}{(x+3)(x+2)(x-4)}$$

$$= \frac{2x^2+5x-3-(x^2+8x+12)}{(x+3)(x+2)(x-4)}$$

$$= \frac{2x^2+5x-3-x^2-8x-12}{(x+3)(x+2)(x-4)} = \boxed{\frac{x^2-3x^2-15}{(x+3)(x+2)(x-4)}}$$

$$\frac{2}{3} - \frac{1}{4}$$

$$\frac{4}{4} \cdot \frac{2}{3} - \frac{1}{4} \cdot \frac{3}{3}$$

$$\frac{4 \cdot 2}{12} - \frac{1 \cdot 3}{12}$$

$$\frac{8}{12} - \frac{3}{12} = \frac{8-3}{12} = \frac{5}{12}$$

$$(x+3)(2x-1)$$

$$= 2x^2 - x + 6x - 3 = 2x^2 + 5x - 3$$

$$(x+6)(x+2)$$

$$= x^2 + 2x + 6x + 12 = x^2 + 8x + 12$$

Definition: A function of the form $f(x) = kx^r$, where k and r are real numbers, is called a _____. We consider several cases:

$r = n$, where n is a positive integer

$r = 1/n$, where n is a postive integer

$r = -1$

Example 5: Write the following in its equivalent radical function format.

(a) $x^{3/5}$ (b) $x^{9/2}$ (c) $(4x+7)^{3/11}$

Notes about finding the domain of a function involving a radical:

- If you are dealing with an even root, the inside of the root cannot be negative. If the even root is in the denominator, then the inside of the root cannot be zero either.

- If you are dealing with an odd root, the inside of the root can be anything. If the odd root is in the denominator, then the inside of the root cannot be zero.

Example 6: Find the domain of the following functions.

a) $g(x) = \sqrt{2x-5}$

b) $h(x) = \sqrt[3]{x-1}$

c) $f(x) = \dfrac{\sqrt{6-5x}}{\sqrt[3]{3x-5}}$

Example 7: Find the domain of $m(x) = \dfrac{\sqrt[8]{10-3x}}{(x-4)(4x-7)}$

$10-3x \geq 0$

$\frac{10}{3} \geq \frac{3x}{3}$

$\frac{10}{3} \geq x$

$\frac{10}{3} = 3.\overline{3}$

$x-4 \neq 0$

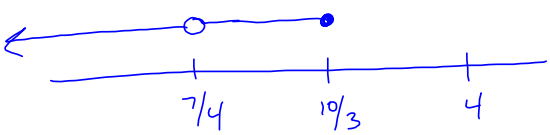
$x \neq 4$

$4x-7 \neq 0$

$\frac{4x}{4} \neq \frac{7}{4}$

$x \neq 7/4$

1.75



1. can't divide by zero
2. can't take the even root of a negative #
3. can only take the log of a positive #

$\sqrt[3]{-8} = -2$ $(-2)(-2)(-2) = -8$

$\cancel{\sqrt[4]{-16}}$ $(4)(4) = 16$

$(-4)(-4) = 16$

$\boxed{(-\infty, 7/4) \cup (7/4, 10/3]}$

Rationalizing the numerator or denominator of a fraction means to get rid of any radicals in that part of the fraction. This often involves multiplying by the conjugate.

Example 8: Rationalize the denominator and simplify the function below:

$$\frac{x-9}{\sqrt{x}+3}$$

The difference quotient is the quantity $\frac{f(x+h)-f(x)}{h}$ and is a very important concept used in calculus.

Example 9: Find and simplify the difference quotient for $f(x) = 3x^2 - 7x + 4$

Example 10: Find and simplify the difference quotient for the following functions:

a) $f(x) = \frac{x-2}{3x-7}$

$$\frac{f(x+h)-f(x)}{h}$$

$$(3x-7)(x+h-2) = 3x^2+3xh-6x-7x-7h+14$$
$$= 3x^2+3xh-13x-7h+14$$
$$(x-2)(3x+3h-7) = 3x^2+3xh-7x-6x-6h+14$$

$$f(x+h) = \frac{x+h-2}{3(x+h)-7} = \frac{x+h-2}{3x+3h-7}$$

$$f(x+h)-f(x) = \frac{x+h-2}{3x+3h-7} - \frac{x-2}{3x-7}$$
$$= \frac{(3x-7)}{(3x-7)}\cdot\frac{(x+h-2)}{(3x+3h-7)} - \frac{(x-2)}{(3x-7)}\cdot\frac{(3x+3h-7)}{(3x+3h-7)}$$
$$= \frac{(3x-7)(x+h-2)-(x-2)(3x+3h-7)}{(3x-7)(3x+3h-7)}$$
$$= \frac{\cancel{3x^2}+\cancel{3xh}-\cancel{13x}-7h+\cancel{14}-(\cancel{3x^2}+\cancel{3xh}-\cancel{7x}-\cancel{6x}-6h+\cancel{14})}{(3x-7)(3x+3h-7)}$$
$$= \frac{-7h+6h}{(3x-7)(3x+3h-7)} = \frac{-h}{(3x-7)(3x+3h-7)}$$

$$\frac{f(x+h)-f(x)}{h} = \frac{\frac{-h}{(3x-7)(3x+3h-7)}}{h}$$
$$= \frac{-\cancel{h}}{(3x-7)(3x+3h-7)}\cdot\frac{1}{\cancel{h}} = \boxed{\frac{-1}{(3x-7)(3x+3h-7)},\ h\neq 0}$$

b) $g(x) = \sqrt{5x-8}$

$$\frac{g(x+h)-g(x)}{h}$$

$$g(x+h) = \sqrt{5(x+h)-8}$$
$$= \sqrt{5x+5h-8}$$

$$= \frac{\left(\sqrt{5x+5h-8}-\sqrt{5x-8}\right)}{h}\cdot\frac{\left(\sqrt{5x+5h-8}+\sqrt{5x-8}\right)}{\left(\sqrt{5x+5h-8}+\sqrt{5x-8}\right)}$$

$$\left(\sqrt{5x+5h-8}-\sqrt{5x-8}\right)\left(\sqrt{5x+5h-8}+\sqrt{5x-8}\right)$$
$$= \sqrt{5x+5h-8}\sqrt{5x+5h-8} + \cancel{\sqrt{5x-8}\sqrt{5x+5h-8}} - \cancel{\sqrt{5x-8}\sqrt{5x+5h-8}} - \sqrt{5x-8}\sqrt{5x-8}$$
$$= (5x+5h-8)^{1/2}(5x+5h-8)^{1/2} - (5x-8)^{1/2}(5x-8)^{1/2}$$
$$= 5x+5h-8-(5x-8)$$
$$= \cancel{5x}+5h\cancel{-8}\cancel{-5x}\cancel{+8} = 5h$$

$$= \frac{5\cancel{h}}{\cancel{h}\left(\sqrt{5x+5h-8}+\sqrt{5x-8}\right)}$$
$$= \boxed{\frac{5}{\sqrt{5x+5h-8}+\sqrt{5x-8}},\ h\neq 0}$$

Definition: Functions whose definition involve more than one rule are called _____.

Example 11: Given $f(x) = \begin{cases} x^2 - 7 & \text{if } x < 3 \\ -2x + 4 & \text{if } x \geq 3 \end{cases}$

a) Find $f(0)$

b) Find $f(3)$

c) Find $f(4)$

d) Sketch a graph of $f(x)$.

Example 12: Find the domain of the following function:

$$g(x) = \begin{cases} \dfrac{5}{x-3} & \text{if } x < -8 \\ \sqrt{5+x} & \text{if } x \geq -8 \end{cases}$$

Example 13: Graph the following piecewise defined function. What is the domain of $f(x)$?

$$f(x) = \begin{cases} -3x^2+2 & \text{if } -2 < x \le 1 \\ 2x-5 & \text{if } 2 \le x < 3 \\ x & \text{if } x \ge 5 \end{cases}$$

$-3x^2+2,\ -2 < x \le 1$

$x=-2 \quad -3(-2)^2+2=-10 \quad (-2,-10)$ open

$x=1 \quad -3(1)^2+2=-1 \quad (1,-1)$ closed

$x=0 \quad -3(0)^2+2=2 \quad (0,2)$ closed

$2x-5,\ 2 \le x < 3$

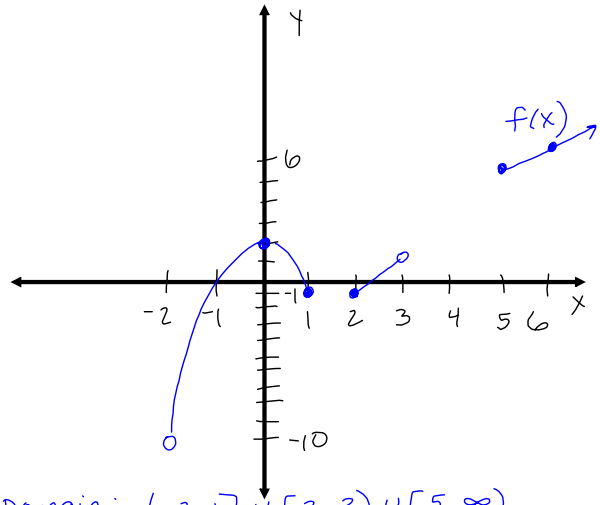
$x=2 \quad 2(2)-5=-1 \quad (2,-1)$ closed

$x=3 \quad 2(3)-5=1 \quad (3,1)$ open

$x,\ x \ge 5$

$x=5 \quad 5 \quad (5,5)$ closed

$x=6 \quad 6 \quad (6,6)$ closed



Domain: $(-2,1] \cup [2,3) \cup [5,\infty)$

Definition: The absolute value function, $f(x) = |x|$, can be writen as a piecewise defined function as given below:

$$f(x) = |x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \ge 0 \end{cases}$$

Example 14: Write $f(x) = |5x-9|$ as a piecewise defined function.

Example 15: A taxi-cab company in a certain town charges all customers a base fee of \$3 per ride. They then charge an additional 50 cents per mile for the first 10 miles traveled and \$1/mile for each mile over 10 miles. Write a piecewise function, $C(x)$, for the cost of a cab ride if x represents the number of miles traveled.

Section 5.2 Highly Suggested Homework Problems: 1, 5, 7, 11, 15, 17, 21, 25, 29, 31, 33, 35, 36, 37, 41, 47, 49, 51, 53

Section 5.3 - Exponential Functions

Properties of Exponents: If a and b are positive numbers and x and y are any real numbers, then

1. $a^0 = 1$
2. $a^{-x} = \dfrac{1}{a^x}$
3. $a^x a^y = a^{x+y}$
4. $\dfrac{a^x}{a^y} = a^{x-y}$
5. $(a^x)^y = a^{xy}$
6. $(ab)^x = a^x b^x$
7. $\left(\dfrac{a}{b}\right)^x = \dfrac{a^x}{b^x}$
8. For $a \neq 1$, $a^x = a^y$ if and only if $x = y$
9. For $x \neq 0$, $a^x = b^x$ if and only if $a = b$

Example 1: Simplify the following and express the answer using no negative exponents.

a) $ab^6(2a^3b^{-3})$

b) $\left(\dfrac{2x^{-2}y^8}{5x^3y^{-2}}\right)^3$

c) $\sqrt[3]{x^{15}y^{-3}}$

Example 2: Simplify the following and express the answer using no negative exponents.

a) $\dfrac{\sqrt[7]{xy^8}\sqrt[2]{x^5y^3}}{x^{-3}y^4}$

$$= \frac{(xy^8)^{1/7}(x^5y^3)^{1/2}}{x^{-3}y^4} = \frac{x^{1/7}\cdot(y^8)^{1/7}\cdot(x^5)^{1/2}\cdot(y^3)^{1/2}}{x^{-3}y^4} = \frac{x^{1/7}y^{8/7}x^{5/2}y^{3/2}}{x^{-3}y^4}$$

$$= \frac{x^{1/7}x^{5/2}}{x^{-3}}\cdot\frac{y^{8/7}y^{3/2}}{y^4} = \frac{x^{1/7+5/2}}{x^{-3}}\cdot\frac{y^{8/7+3/2}}{y^4}$$

$$= \frac{x^{37/14}}{x^{-3}}\cdot\frac{y^{37/14}}{y^4} = x^{37/14+3}\,y^{37/14-4}$$

$$= x^{79/14}\,y^{-19/14}$$

$$= \boxed{\frac{x^{79/14}}{y^{19/14}}}$$

b) $\dfrac{(3x^{-3}y^4z^9)^{-2}}{(4xy)^3x^3y^{-8}}$

$$= \frac{3^{-2}(x^{-3})^{-2}(y^4)^{-2}(z^9)^{-2}}{4^3x^3y^3x^3y^{-8}}$$

$$= \frac{3^{-2}x^6y^{-8}z^{-18}}{4^3x^6y^{-5}} = \frac{3^{-2}}{4^3}\cdot\frac{x^6}{x^6}\cdot\frac{y^{-8}}{y^{-5}}\cdot z^{-18}$$

$$= \frac{1}{576}\cdot 1\cdot y^{-8+5}z^{-18} = \frac{1}{576}\cdot y^{-3}z^{-18}$$

$$= \boxed{\frac{1}{576y^3z^{18}}}$$

Definition: The **exponential functions** are the functions of the form $f(x) = a^x$, where the base a is a positive constant with $a \neq 1$.

Properties of the Graphs of $f(x) = a^x$

1. Domain is the set of all real numbers.
2. Range is the set of all positive real numbers.
3. All graphs pass through the point $(0, 1)$.
4. The graph is continuous (no holes or jumps).
5. The x axis is a horizontal asymptote (but only in one direction).
6. If $a > 1$, the graph is increasing (exponential growth).
7. If $0 < a < 1$, the graph is decreasing (exponential decay).

Example 3: Graph $f(x) = 2^x$, $g(x) = 5^x$, $h(x) = \left(\frac{1}{2}\right)^x$, and $k(x) = \left(\frac{1}{5}\right)^x$.

Definition: The most common base is the number e.

Example 4: Determine the domain of each function below:

a) $f(x) = 3^{x-2}$

b) $g(x) = 7^{\sqrt[3]{x-2}}$

c) $h(x) = 5^{\frac{x-2}{x}}$

Example 5: What is the domain of $k(x) = \frac{\sqrt[2]{5x-9}}{3^x}$?

① Can't take the even root of a negative #

② Can't divide by zero

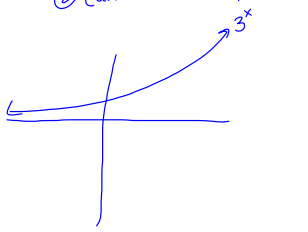
$5x - 9 \geq 0$

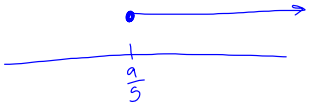
$\frac{5x}{5} \geq \frac{9}{5}$

$x \geq 9/5$

$3^x \neq 0$

3^x is never zero





$\boxed{[9/5, \infty)}$

Example 6: Solve each equation for x:

a) $9^{x-1} = 3^{1+x}$

b) $x^2e^x - 5xe^x = 0$

Example 7: Solve the equation for x: $64^{3x-9} = 256^{5x-2}$

$(4^3)^{3x-9} = (4^4)^{5x-2}$

$4^{3(3x-9)} = 4^{4(5x-2)}$

$4^{9x-27} = 4^{20x-8}$

$\cancel{9x} - 27 = 20x - \cancel{8}$

$-\cancel{9x} + 8 \quad -9x \quad \cancel{+8}$

$\frac{-19}{11} = \frac{11x}{11}$

$\boxed{-19/11 = x}$

Example 8: The population of a particular town increases by 5% each year. It's population after t years is given by $P(t) = 3000(1.05)^t$ where t is the number of years since 1980. What will the population be in 2018?

Continuously Compounded Interest: If $\$P$ is invested into an account paying continuously compounded interest over a time period of t years, the amountin the account at the end of the time period (A) is given by

where r=annual interest rate (as a decimal).

Example 9: What amount will an account have after five years if $1,000 is invested at an annual rate of 3.25% compounded continuously?

Example 10: John needs $25,000 five years from now to buy a car. How much does he need to invest in his account today if his account pays interest at an annual rate of 1.75% compounded continuously?

$A = 25000$
$P = ?$
$r = .0175$
$t = 5$

$A = Pe^{rt}$

$25000 = Pe^{.0175 \cdot 5}$

$25000 = 1.091442264P$

$\frac{25000}{1.091442264} = P$

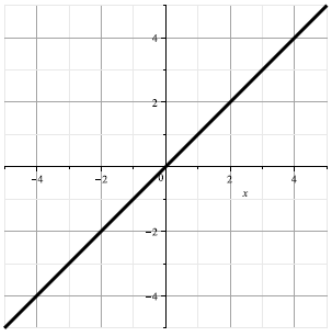
$\boxed{\$22905.47 \approx P}$

Section 5.3 Highly Suggested Homework Problems: 1, 5, 9, 15, 19, 23, 27, 31, 35, 37, 41, 45, 47

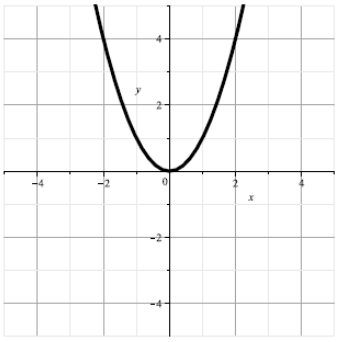
Section 5.4 - Combinations of Functions

Example 1: Graph the following functions on the appropriate graph below and describe the transformation.

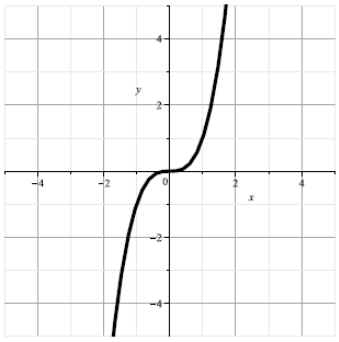
a) $g(x) = x + 4$ b) $g(x) = (x-3)^2$ c) $g(x) = -4x^3$



$f(x) = x$



$f(x) = x^2$



$f(x) = x^3$

Transformations

Vertical and Horizontal Shifts: Suppose $c > 0$. To obtain the graph of

- $y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units _____.
- $y = f(x) - c$, shift the graph of $y = f(x)$ a distance c units _____.
- $y = f(x - c)$, shift the graph of $y = f(x)$ a distance c units _____.
- $y = f(x + c)$, shift the graph of $y = f(x)$ a distance c units _____.

Vertical Stretching and Reflecting: Suppose $c > 1$. To obtain the graph of

- $y = cf(x)$, _____ the graph of $y = f(x)$ _____ by a factor of c.
- $y = (1/c)f(x)$, _____ the graph of $y = f(x)$ _____ by a factor of c.
- $y = -f(x)$, reflect the graph of $y = f(x)$ about the _____.

Example 2: Give the appropriate parent function and the list of transformations needed to graph the functions below.

a) $f(x) = |x-2|+3$

b) $g(x) = -2(x-3)^2$

c) $h(x) = 4e^{x+2}+6$

Example 3: If $g(x) = -2f(x+1)-3$, describe how $f(x)$ was transformed to get $g(x)$.

- shift the graph 1 unit to the left
- reflect the graph about the x-axis
- stretch the graph vertically by a factor of 2
- shift the graph 3 units down

Example 4: If the graph of $f(x) = x^3$ is shifted left 2 units, vertically expanded by a factor of 5, reflected across the x-axis, and then shifted up 3 units, what is the equation of the resulting graph?

Combinations of Functions: Let f and g be two functions. Then, the following functions can be defined:

- $(f+g)(x) = f(x) + g(x)$
- $(f-g)(x) = f(x) - g(x)$
- $(fg)(x) = f(x)g(x)$
- $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ where $g(x) \neq 0$

Example 5: Given $f(x) = \sqrt{2x+5}$ and $g(x) = 3x^2 + 7$, find

a) $(f+g)(x)$

b) $(f-g)(x)$

c) $(fg)(x)$

d) $\left(\frac{f}{g}\right)(x)$

Definition: Given two functions f and g, the **composite function** $f \circ g$ (also called the **composition** of f and g) is defined by

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f.

Example 6: Let $f(x) = x^{10}$, $g(x) = 2x^2 - 5x + 3$, $h(x) = \sqrt{x}$, and $k(x) = 2^x$, find the following:

a) $f(g(x))$

b) $(h \circ g)(x)$

c) $k(g(x))$

d) $f(g(3))$

e) $h(f(4))$

Example 7: Use the table below to find the value of each expression.

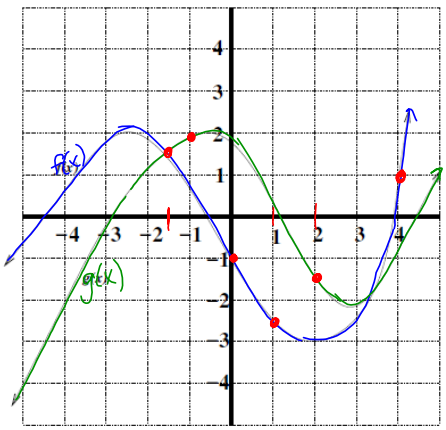
x	1	2	3	4	5	6
$f(x)$	4	1	8	7	4	9
$g(x)$	6	9	1	10	5	3

a) $f(g(3))$

b) $g(f(5))$

c) $(f \circ g)(6)$

Example 8: Use the given graphs of f and g to estimate the value of each expression.



a) $f(g(2)) = f(-1.5) = \boxed{1.5}$

b) $g(f(0)) = g(-1) = \boxed{1.9}$

c) $(f \circ f)(4) = f(f(4)) = f(1) = \boxed{-2.5}$

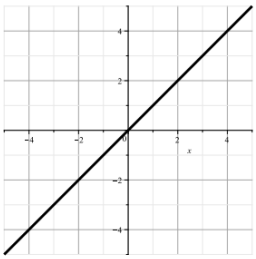
Section 5.4 Highly Suggested Homework Problems: 1, 3, 7, 9, 13, 15, 19, 23, 27, 31, 33, 37, 43, 47

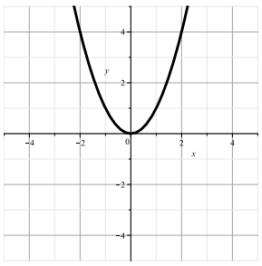
Section 5.5 - Logarithms

Definition: A function f is said to be _____ if each range value corresponds to exactly one domain value.

Horizontal Line Test: If every horizontal line intersects the graph of a function f in no more than one place, then f is a one-to-one function.

Example 1: Which of the following functions are one-to-one?





Definition: If f is a one-to-one function with domain A and range B, then its **inverse function** has domain B and range A. Thus, if (a,b) is a point on the graph of f, then (b,a) is a point on the graph of the inverse of f. Thus, if f is one-to-one, then the graph of its inverse is a reflection about the line $y = x$.

Example 2: Sketch a graph of 2^x and its inverse.

Definition: The inverse of an exponential function is called a **logarithmic function**. For $b > 0$ and $b \neq 1$,

$$y = \log_b x \quad \Leftrightarrow \quad b^y = x$$

Example 3: Solve for x, y, or b without a calculator.

a) $\log_3 x = 2$

b) $\log_b e^{-4} = -4$

c) $\log_{49}(\frac{1}{7}) = y$

Properties of Logarithmic Functions

If b, M, and N are positive real numbers, $b \neq 1$, and x is a real number.

1. $\log_b 1 =$
2. $\log_b b =$
3. $\log_b b^x =$
4. $b^{\log_b x} =$, $x > 0$
5. $\log_b MN = \log_b M + \log_b N$
6. $\log_b \frac{M}{N} = \log_b M - \log_b N$
7. $\log_b M^N = N \cdot \log_b M$
8. $\log_b M = \log_b N$ if and only if $M = N$

Example 4: Use properties of logarithms to fully exapnd the expression below into a sum or difference of simpler logarithms.

$\log_3 \left(\frac{z^2+4}{x^2 y} \right)$

Example 5: Given $\log_b 3 = 0.6826$ and $\log_b 4 = 0.8614$, find the following:

a) $\log_b 48$

$= \log_b (4 \cdot 4 \cdot 3)$

$= \log_b (4) + \log_b (4) + \log_b (3)$

$= .8614 + .8614 + .6826 = \boxed{2.4054}$

$\rightarrow \log_b MN = \log_b M + \log_b N$

$\rightarrow \log_b \frac{M}{N} = \log_b M - \log_b N$

$\rightarrow \log_b M^N = N \cdot \log_b M$

b) $\log_b \left(\frac{16}{b^2} \right)$

$= \log_b 16 - \log_b b^2$

$= \log_b 4^2 - 2 \log_b b$

$= 2 \log_b 4 - 2(1)$

$= 2(.8614) - 2(1) = \boxed{-.2772}$

$b^? = b$

$b^1 = b$

Special Logarithms

- Common Logarithm

- Natural Logarithm

Change of Base Formula:

$\log_a x = \frac{\log_b x}{\log_b a}$

Example 6: Evaluate $\log_8 17$.

Example 7: Find the domain of $f(x) = \dfrac{\sqrt[7]{2x-9}}{\log_2(4x-1)}$.

Example 8: Find the domain of $f(x) = \dfrac{\ln(-4x+7)}{\sqrt[5]{2x+9}}$.

① can't take the even root of a negative number
② Can't divide by zero
③ can only take the log of positive #'s

$\sqrt[5]{2x+9} \neq 0$

$\left((2x+9)^{1/5}\right)^5 \neq (0)^5$

$2x+9 \neq 0$

$\frac{2x}{2} \neq \frac{-9}{2}$

$x \neq -9/2$

$-4x+7 > 0$

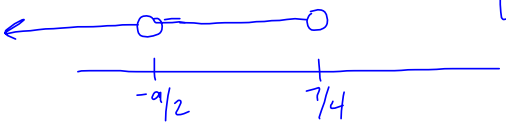
$\frac{7}{4} > \frac{4x}{4}$

$7/4 > x$

OR

$\frac{-4x}{-4} > \frac{-7}{-4}$

$x < 7/4$



$\boxed{(-\infty, -9/2) \cup (-9/2, 7/4)}$

Example 9: Solve the following for the exact value of x:

a) $\log_b x = \frac{2}{3}\log_b 27 + 2\log_b 2 - \log_b 3$

b) $\log x + \log(x-3) = \log 10$

c) $3^{2x-1} - 7 = 5$

Example 10: Solve the following for the exact value of x.

a) $\log_3(x+7) = \log_3(2x) + 4$

$\log_3(x+7) - \log_3(2x) = 4$

$\log_3\left(\frac{x+7}{2x}\right) = 4$

$3^4 = \frac{x+7}{2x}$

$\frac{81}{1} = \frac{x+7}{2x}$

$2x(81) = x+7$

$162x = x+7$

$-x \quad -x$

$\frac{161x}{161} = \frac{7}{161}$

$\boxed{x = 7/161 = 1/23}$

Check:

$\log_3(1/23 + 7) = \log_3(2 \cdot 1/23) + 4$ ✓

b) $\frac{4^{5x}}{2^{-3x}} = 16^{2x} \cdot 5$

$\frac{(2^2)^{5x}}{2^{-3x}} = (2^4)^{2x} \cdot 5$

$\frac{2^{10x}}{2^{-3x}} = 2^{8x} \cdot 5$

$2^{10x+3x} = 2^{8x} \cdot 5$

$\frac{2^{13x}}{2^{8x}} = \frac{\cancel{2^{8x}} \cdot 5}{\cancel{2^{8x}}}$

$\frac{2^{13x}}{2^{8x}} = 5$

$2^{13x-8x} = 5$

$2^{5x} = 5$

$\ln(2^{5x}) = \ln(5)$

$\frac{\cancel{5}x\cancel{\ln(2)}}{\cancel{5}\cancel{\ln(2)}} = \frac{\ln(5)}{5\ln(2)}$

$\boxed{x = \frac{\ln(5)}{5\ln(2)}}$

Example 11: How long will it take for the amount in an account to double if the money is compounded continuously at an interest rate of 2.5% per year?

Section 5.5 Highly Suggested Homework Problems: 1, 5, 9, 13, 17, 25, 29, 33, 37, 41, 49, 53, 59, 63